MCS-331	REAL TIME SYSTEMS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	Syllabus deals with issues in real time operating systems
CO2	Importance of deadlines and concept of task scheduling
CO3	Student will be able to understand and design real time operating systems which are backbone of embedded industry
CO4	Student will be able to understand Common Approaches to Real Time Scheduling
CO5	To learn about RAC

MODULE I

Introduction: Concept of Real Time System, Issues in real time computing, Performance measures of Real Time System, Issues in Real Time Computing, Performance measures of Real time Systems, Real Time Application.Task Assignment and Scheduling: Different task model, Scheduling hierarchy, offline vs Online Scheduling, Clock Drives.

Model of Real Time System: Processor, resources, temporal parameter, Periodic Task Model, Sporadic Task Model, Precedence Constraints and Data Dependencies, Scheduling hierarchy Scheduling of Periodic Task: Assumptions, fixed versus dynamic priority algorithms, schedulability test for fixed priority task with arbitrary deadlines.

MODULE II

Scheduling of A periodic and Sporadic Tasks: Assumptions and approaches, deferrable, sporadic servers, slack stealing in deadline driven and fixed priority systems. Two level scheme for integrated scheduling, Scheduling for applications having flexible constrains.

MODULE III

Resources and Resource Access Control: Assumptions on resources and their usage, resource contention, resource access control(Priority Ceiling Protocol, Priority Inheritance protocol, Slack Based Priority Ceiling Protocol, Peremption Ceiling Protocol). Multi Processor Scheduling: Model of multi processor and distributed systems, Scheduling algorithms for end to end periodic tasks in homogeneous/heterogeneous systems, Predictability and validation of dynamic multiprocessor system. Real time Communication: Model of real time Communication, Priority base service

For switched network, Weighted Round Robin Service, Medium access Control Protocol, Real Time Protocol.

Books and References:

1. Jane .W. S. Liu Real Time Systems Pearson Education.

2. Krishna .C.M Real Time Systems Mc-Graw Hill Publication.

Course Outcome:

CO1	To present the mathematical model of the system.
CO2	To develop real-time algorithm for task scheduling.
CO3	To understand the working of real-time operating systems and real-time database
CO4	To work on design and development of protocols related to real-time communication.
CO5	Able to implement RAC